

The minimal effects of body-worn camera adoption on organizational outcomes*

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Abstract

Research Summary

Body-worn cameras (BWCs) were among the most rapidly and widely adopted policing technologies during the 2010s. Although BWCs were introduced largely with the expectation of reducing police misconduct and excessive use of force, decades of research have produced mixed evidence regarding their effects on officer behavior. We argue that BWCs may also influence organizational outcomes by increasing accountability through enhanced opportunities for retrospective review. Drawing on a nationally representative survey of police agencies and administrative records, we examine whether BWC adoption affects officer separations and police settlements. Leveraging the staggered adoption of BWCs across agencies, we estimate treatment effects using heterogeneity-robust difference-in-differences estimators. We find little evidence that BWC adoption produces meaningful changes in officer separations or police settlements. Although BWC adoption may increase the transparency and observability of individual officer behavior, these benefits do not necessarily translate into broader organizational consequences.

Policy Implications

Given the substantial costs associated with adopting, operating, and maintaining BWC programs, policy-makers and practitioners should be cautious about expecting BWCs to substantially improve workforce stability or reduce civil liability beyond their role in enhancing transparency and oversight. Rather, decisions to adopt and invest in BWCs should be based on a realistic understanding of what these technologies can and cannot be expected to accomplish as well as the organizational and implementation conditions under which their intended effects are most likely to emerge.

Keywords: Body-worn camera, officer separation, police settlement

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1 Introduction

During the 2010s, several high-profile police killings of citizens generated widespread calls for improved officer accountability. In response, law enforcement agencies adopted a range of new policies and practices. First among equals was the adoption of body-worn cameras (BWCs). These were deployed by many police agencies with the expectation that their presence during police-civilian interactions would reduce the use of deadly force by police officers. In a country not known for the rapid diffusion of policing practices, given the highly decentralized organization of policing in the U.S., the pace of adoption of BWC was notable. By 2016, 80 percent of large police departments adopted BWCs (Hyland, 2018). By 2020, 100 percent of large police departments had done so and smaller agencies were following suit (Goodison and Brooks, 2023).

This remarkably rapid adoption of BWCs has resulted in a substantial empirical literature evaluating the effects of their deployment on the use of force by officers (Lum et al., 2020; Kim, 2025; Miller and Chillar, 2022). Interestingly, given the initial positive results found in several early RCTs, the current scholarly consensus is that the effects of BWCs are quite mixed (Lum et al., 2020). Although the reasons for these inconsistent results are not yet fully understood, this variability can be attributed to underpowered research designs in many studies caused by a sparse outcome measure at the agency level, heterogeneity in the BWC recording policies, and differences in footage management and review policy between law enforcement agencies.

The mixed findings on the effects of BWC on decision making have left the retrospective review as the primary mechanism for BWC to potentially enhance police accountability and transparency, rather than through the anticipatory behavioral effect. That is, one of the most straightforward ways in which BWCs can accomplish this is by enabling retrospective review of officer compliance with police regulations and relevant statutory laws by police managers, thereby facilitating the identification and correction of officer misconduct.

Although the existence of this post-hoc review capability creates an incentive for better compliance with relevant policies, it may also increase an incentive for turnover if either the first incentive is too small or the prospect of review and macro- or micro-management is too great. In this sense, enhanced police accountability through the retrospective review capacity may be, at least in part, reflected in workforce outcomes such as officer separations following the adoption of BWCs. In addition, a reviewable recording of officer behavior has the potential to alter the likelihood of civilian complaints, civil lawsuits, and police settlements as people outside the police such as attorneys, prosecutors or the complainants may gain greater opportunities to review and use it for complaints. Consistent with this perspective, recent

scholarship has highlighted the potential for BWCs to have sizable effects on settlements of police organizations even the uncertain effects of BWCs on individual officer behavior (Powell, 2024). This possibility underscores the importance of examining the managerial and organizational consequences of BWC adoption. If BWCs primarily reshape accountability through enhanced observability and retrospective review, their most consequential effects may emerge through organizational processes and outcomes rather than through immediate changes in officer behavior during police-citizen encounters.

In this study, we estimate the impact of body worn cameras on officer separation and police settlement. Both outcomes have important implications for how the introduction of new technology affects the management and operation of police organizations by increasing accountability. Drawing on representative administrative survey data, this study exploits the staggered adoption of BWCs across police departments to identify the effects of BWCs on the likelihood of officer separation and, for a subset of jurisdictions, the effects of BWCs on the likelihood and dollar value of police settlements. For the separation analysis, we rely on three large states where 353 out of 567 agencies reported the adoption of camera as of 2016. For the settlement analysis, we use data from the Washington Post covering 20 large police agencies in the US. In combination, these three data sources and a staggered-diff-in-diff research design produces a causal result comparing agencies that deployed body worn cameras and the comparison agencies that either did not deploy them or did so somewhat later.

We find little evidence that the adoption of BWCs affected overall probability of officer separation in three states, allowing us to rule out all but modest sized effects. The small and somewhat fragile observed average treatment effect appears to vary in magnitude across jurisdictions and also across adoption timing. However, even these significant sub-group effects appear temporary and dissipate by the end of our follow-up period. With regards to settlement outcomes, we find little evidence that BWC adoption affects the yearly settlement count in main specification, while an alternative estimator using a less granular measure of treatment provides some evidence of a reduction in settlement counts. For payment amount, we do not find evidence that annual total payment changed as a result of BWC adoption regardless of the estimators. Taken together, the findings indicate that the consequences of BWC adoption may often be null, or appear to be fragile and non-uniform. This suggests, consistent with earlier studies of the effects of BWCs on office use of force, that BWCs may influence organizational processes in ways that vary across agencies, officer experience, and outcome domains. Such heterogeneity may explain why the average effects of BWCs on separations and settlements are often statistically indistinguishable from zero.

This paper proceeds with the following steps – In the background section, we describe the conceptualization of organizational outcomes in police agency, and the relevance of BWCs with separation and

settlement outcomes. Then, we discuss the data sources and empirical strategy. In the result section, we report the main findings. Finally, the implications for theory and policy are discussed.

2 Prior Literature

2.1 Polices and practices that shape officer and organizational management

After BWC usage rapidly expanded in the 2010s (Hyland, 2018), empirical studies began scrutinizing the effects that these early BWC deployments had on officer use of force (Lum et al., 2020; Braga et al., 2020, 2022; Miller and Chillar, 2022; Kim, 2025). These studies, most examining individual police agency deployments through a mixture of small-scale randomized control trials and other analytical approaches, reported that BWC use led to sizable reductions in officer use of force (Ariel et al., 2015; Henstock and Ariel, 2017; Braga et al., 2017). Despite this promising start, the sparseness of the outcome measure, usually use of deadly force, and the small size of many jurisdictions studied, has contributed to scholarly concern that these estimates might not represent the average effects when cameras were deployed across a wide variety of police organizations (Lum et al., 2019, 2020). However, support for these early results has been bolstered by more recent national studies which have also reported reductions in deadly use of force in adopting jurisdictions (Kim, 2025; Miller and Chillar, 2022).

Despite these lingering uncertainties, the use and analysis of BWC has continued, and scholars have examined a sizable number of additional outcomes under the assumption that the presence of BWC cameras on police officers could impact a range of incident-related outcomes. These include the perception of BWCs among officers (Adams and Mastracci, 2019; Jennings et al., 2014), (White et al., 2018), officer compliance with BWC policies (Braga et al., 2022), officer safety (Ariel et al., 2016), as well as officer proactivity (Lawrence and Peterson, 2020). Most of these studies have not yet reported a consistent statistically significant impact on most measures (Lum et al., 2019, 2020). The one exception to this is the impact of BWC on citizen complaints, where studies have consistently documented that BWCs reduce citizen complaints (Ariel et al., 2015; Hedberg et al., 2017; Braga et al., 2018, 2022; Lum et al., 2020).

While behavioral or decision-making changes in officer-citizen encounters have been the primary outcome of interest in the field, scholars have also begun to explore the ways in which BWC may change downstream criminal justice practices. The earliest of these studies examined how the availability of video recordings during domestic violence calls for service could change case outcomes not only during the incident itself but also once the resulting case was sent for prosecutorial and judicial review (Morrow et al., 2016). While limited in scope, this early work suggested that standard case processing activities

of charging, felony review, filing, and discovery all had the potential to be impacted by the availability of BWC footage recorded during police response to calls for service. More recently, [Powell \(2024\)](#) assessed the impact of the deployment of BWCs on organizational outcomes, finding that BWC deployment led to large changes in the probability and amount of police settlements for agencies facing civil lawsuits. These results, taken together, provide early evidence that the most significant and sustained impacts of BWC could be beyond the instantaneous decision-making changes that scholars have most extensively examined and instead lie in the retrospective transparency of BWC.

As the retrospective transparency provides a new opportunity to check the validity of alleged officer misconduct, influencing how police agencies manage their workforce and civilian complaints, the effect of BWCs may often be reflected in organizational outcomes. Organizational outcomes can be conceptualized as the characteristics that reflect the functioning of institutions. These encompass a wide range of institutional aspects, including performance, culture, managerial systems, workforce dynamics, and organization’s relationship with the public. In police agencies, some of the dimensions can be manifested through specific measures. For example, a clearance rate reflects, in part, the performance of a police agency. Also, officer injury in the line of duty represents workforce safety. Managerial systems can be reflected in officer discipline or dismissal processes. Citizen complaints and police settlements indicate organizational accountability as well as the relationship between an institution and the public. Organizational outcomes are distinguished from the individual-level outcomes, which are often the consequences of direct individual behavior. However, this does not mean that organizational outcomes are unrelated to individual-level outcomes. To a varying degree, organizational outcomes often reflect both the aggregated individual actions and institutional systems. Some organizational outcomes like clearance reflect the combination of individual officer’s performance and organizational policy, while others like settlement and retention are largely driven by institutional governance systems beyond immediate individual actions.

Among these various dimensions, this study focuses on officer separation and police settlement. These two outcomes reflect how police agencies monitor, evaluate, and respond to alleged officer misconduct. These outcomes are often understudied because organizational data are often difficult to obtain. This study combines multiple datasets that contain information about separation, settlement, and BWC adoption among police agencies in US. The remaining part of this section provides the general landscape of officer separation and police settlements and how BWCs are related to these two outcomes.

2.2 BWCs and officer separation

The staffing crisis in U.S. law enforcement has been increasingly identified as a major concern in policing profession. Media reports and empirical studies documented substantial increases in officer separation beginning in 2020 (Mourtgos et al., 2022; Wallace, 2020; Bailey, 2020). For example, Mourtgos et al. (2022) reported 279% increase in voluntary resignation following the death of George Floyd in 2020 among large police departments relative to its counterfactual in the western United States. While Mourtgos et al. (2022) suggested the staffing crisis is particularly salient following 2020, the staffing shortages had emerged even before 2020 (Grunwald, 2024; PERF, 2019). Prior research identified general reasons for the officer separation or turnover intention. Commonly identified factors include political activism against excessive use of force (Mourtgos et al., 2022), poor working conditions (Wallace, 2020; Quick and Wolff, 2025), financial factors such as compensation and salary levels (Schuck and Rabe-Hemp, 2018; Quick and Wolff, 2025).

Adopting BWCs may have resulted in unintended consequences of officer separation. Studies showed that deploying BWCs is associated with the increase in burnout and decrease in perceived organizational support, although the perception of BWCs is generally positive among law enforcement officers (Adams and Mastracci, 2019; Jennings et al., 2014). As noted in Hoogesteyn et al. (2025), workplace wellness and organizational support remain crucial components to enhance officer retention. Considering this connection between BWC deployment and officer satisfaction at work, strict activation requirements or policies permitting the intensive supervisory review of BWC footage may increase officers' perception of organizational surveillance, and contribute to workplace stress and dissatisfaction. Associated with this, some police unions opposed the implementation of BWCs, claiming that their adoption should be negotiated through collective bargaining agreement (Bureau of Justice Assistance, 2015; Garcia, 2018). As of December 2023, police union agreements in four states and 10 municipalities include provisions as to the use of BWCs¹. If the deployment of BWCs disincentivized frontline officer's morale in some manner, BWCs might have functioned as a tool merely for organizational surveillance, worsening the retention problem.

Also, officer separation occurs involuntarily through dismissal. The termination of employment is one of the most punitive forms of officer discipline, while less severe type of disciplinary actions includes written reprimands, demotions, and suspensions (Rushin, 2021). Therefore, the dismissal decision requires sufficient evidence to be established. One of the difficult parts of traditional system without BWCs to validate the officer misconduct lies on the absence of neutral and retrievable evidence that captures the

¹https://ballotpedia.org/States_and_cities_with_police_union_agreements_that_have_provisions_related_to_the_use_of_body-worn_cameras_or_body-worn_camera_video_evidence

alleged misconduct of officers. The recordings documented with BWCs have the potential to play a role here, coupled with other recording devices such as cell phones, dash cams in motor vehicle, and CCTVs. That is, police agencies that adopted BWCs gain capacity of post-hoc evaluation of officer behaviors. With the use of BWC footage policy in the assessment of officer misconduct, officer dismissals decisions may be significantly influenced by the availability of BWC footage. On the one hand, police dismissal potentially would have increased immediately after the adoption of BWCs as supervisors observe pre-existing misconduct by individual officers that were less likely to be known otherwise. On the other hand, some disciplinary actions that used to rely on the subjective judgment of supervisors may be more likely dismissed by the evidence proving the officer’s innocence, resulting in fewer involuntary separations. Importantly, all these hypothetical scenarios assume the actual use of BWC footage by supervisors or managers with enough intensity such that it influences officer separation either voluntarily or involuntarily. If police agencies do not utilize BWCs in the evaluation of officers in fact, the adoption of BWCs might hardly affect the dismissal decisions.

2.3 BWCs and Police Settlement

A police settlement is a binding agreement that resolves the claim about police misconduct raised by civilians without engaging in a trial. Once settled, this often results in monetary compensation to a claimer. *The Washington Post* released an article as to police settlement of 25 large agencies in the US, suggesting that thousands officers were involved with misconduct, costing more than \$3 billion for about 40,000 payments between 2010 and 2020 (Alexander et al., 2022). The settlement costs are paid through private insurance, law enforcement agencies, and a wide range of budgetary arrangements by city, county, and state (Schwartz, 2016). Forsyth et al. (2026) raised a question of what percentage of cases are settled. They examined a sample of 260 notice of claims against Rochester Police Department (RPD) between 2000 and 2012. 89 of 260 claims were settled, approximating 34 percent settlement rate.

Understanding settlement remains crucial given the substantial amount of payment that taxpayers are burdened with. Also, settlement can be represented as a proxy for serious police misconduct (Powell, 2024). Determining what increases or reduces it would help efficient budget managing and police legitimacy. In traditional police organizations, however, decisions regarding settlement often relied on participant or eyewitness testimony and police reports, which could be inaccurate and/or incomplete.

In this regard, the adoption of BWCs may influence police settlement by increasing the availability of objective evidence regarding alleged misconduct. Video footage can help legal actors assess the credibility of the claims, determine whether a claim should be settled, and evaluate the appropriate amount of com-

pensation when a settlement occurs. Due to this increased transparency and accountability mechanisms, to an extent that BWCs enhance officer-civilian interactions and reduce incidents of misconduct, they may also decrease the number of complaints and lawsuit brought against police agencies in the first place (Miller et al., 2014; Police Executive Research Forum, 2018), resulting in fewer settled cases.

At the same time, BWCs help identify how existing claims are assessed and resolved. Footage can help identify unsubstantiated allegations, discouraging weak claims and reducing the likelihood of settlement (Police Executive Research Forum, 2018). Conversely, when misconduct occurred, video evidence may strengthen the credibility of the claims, and increase the likelihood that agencies choose to settle. As a result, BWCs may increase or decrease settlement frequency. Because BWCs may simultaneously enhance the resolution of valid claims while screening out unsubstantiated ones, the overall effect on settlements is theoretically ambiguous and may be attenuated by offsetting mechanisms.

Despite the intriguing possibility that BWCs affect settlement dynamics, the existing literature assessing this possibility consists of descriptive analyses or before-and-after comparisons. Police Executive Research Forum (2018) descriptively compared the number of law suits and the dollar costs of settlement before and after the deployment of BWCs in police agencies in three cities. PERF reported that while lawsuit frequency increased in Mesa and declined in Phoenix, settlement payments decreased in Mesa and Phoenix but increased in Dallas. These results suggest that the consequences of BWC deployment may not be uniform. Powell (2024) used a before-and-after research design to estimate the effect of BWC adoption on settlements. Drawing on data from *Washington Post*, the study constructed panel data of 19 large police agencies between 2010 and 2019. The resulting findings suggested that BWCs decreased police settlements in the years after adoption. Specifically, the deployment of BWCs substantially reduced settlement outcomes, decreasing both the number of police settlements and the median payment amount by approximately 40 percent.

Using the *Washington Post* data for settlement, the present study revisits this question using a difference-in-differences research design that provides a stronger causal estimate of the effect of BWC adoption on settlement. We also present additional analyses to see the robustness of the results.

3 Data

LEMAS-BWC 2016

In order to examine the impact of BWCs on organizational outcomes, we draw on three principal datasets. First, we use the 2016 Law Enforcement Management and Administrative Statistics Body-Worn Camera

Supplement (LEMAS-BWCS) where 3,928 sample police agencies in the US were surveyed on a series of questions about the BWC and its operation. Only a subset of the full sample in LEMAS is drawn as the coverage of each data source is different. To identify the timing of BWC adoption, we use the question that asked the year and month of BWC adoption for the responding agencies. We also include the agencies that did not adopt BWCs by the time of the survey response to construct a comparison pool. The information of some agencies with missing values of BWC adoption timing was searched and sourced from available media reports on the web². Furthermore, the data includes the number of full-time sworn officers for individual agencies. We employ this as a proxy to calculate the settlement rate per officer.

IPED

Table 1: Separation Descriptive Statistics

	Full Sample			California		Florida		Texas	
	Full	Treated	Control	Treated	Control	Treated	Control	Treated	Control
<i>Sample Composition</i>									
N Agencies	567	353	214	102	59	61	79	190	76
N Agency-Months	88,452	55,068	33,384	15,912	9,204	9,516	12,324	29,640	11,856
Avg. Sworn Officers (2016)	239	251	218	350	477	313	174	179	62
<i>Monthly Separations</i>									
Total Separations	1.68	1.71	1.62	1.88	2.13	1.67	1.07	1.64	1.81
Long-Tenure (≥ 10 Years)	0.79	0.77	0.82	0.70	0.66	0.70	0.52	0.83	1.25
Short-Tenure (< 10 Years)	0.89	0.94	0.81	1.17	1.46	0.98	0.55	0.80	0.56
Pre-Treatment Mean	1.66	1.66	—	1.85	—	1.66	—	1.56	—
<i>BWC Adoption Cohorts (N Agencies)</i>									
Early Adopters (≤ 2012)	61	61	0	23	0	6	0	32	0
Mid Adopters (2013–2014)	99	99	0	25	0	20	0	54	0
Late Adopters (≥ 2015)	193	193	0	54	0	35	0	104	0
Never Treated	214	0	214	0	59	0	79	0	76

Separation data was obtained from the Interstate Police Employment Database (IPED) (see [Grunwald 2024](#)). The IPED records job stints of law enforcement officers with person and agency identifier, employment start date, and separation date, although the exact variable names vary across the states. The current study focuses on three large states where the adoption timing of BWCs among police agencies varies. While the oldest IPED observation dates back to the 1900s in terms of employment start date, we restrict the sample to the separation that occurred within 2005-2017. This is because the reliability of IPED has improved in more recent decades. Also, LEMAS survey was conducted in 2016, and it is less clear whether those who did not adopt BWCs by then introduced BWCs afterwards. This restriction resulted

²BWC adoption timings of Boston, Jacksonville, and Prince George police departments were searched through news articles.

in 314,711 job stints for 233,875 officers in the sample states.³ Using the agency name, the separation data was then matched to LEMAS-BWCS 2016 data where it documented the BWC adoption timing for the sample agencies. Because LEMAS covered a limited number of agencies of each state, the matching procedure yielded 148,628 job stints (separations) of 133,458 officers among 567 agencies between 2005 and 2017.

Table 1 reports the descriptive summary of the BWC adoption and separation data. There were 353 agencies that adopted BWCs during the study period while 214 agencies never adopted the camera at least as of 2016. The year 2015 was when the vast majority of agencies adopted BWCs, and the earliest adoption was in 2006 and the latest adoption was in 2016. Table 1 also documents the monthly separation count on average. There was minimal difference in separation count between the BWC agencies (1.71) and No-BWC agencies (1.68) on average. During the pre period, the average separation counts are slightly smaller than the treatment periods across the jurisdictions. Average monthly total separation count (not shown in the table) is highest in California (1.97), followed by Texas (1.68) and Florida (1.33). To see whether the separation changes for experienced officers and relatively younger officers, two additional outcomes are tested separately – (1) a separation of officers whose cumulative employment years are less than 10 years, (2) a separation of officers whose cumulative employment years are 10 or more than ten years. With regards to BWC adoption timing, vast majority of agencies adopted BWCs after 2014 (n= 193), which are classified late adopters, then followed by mid adopters (n= 99) who adopted BWCs between 2013 and 2014, and early adopters (n= 61) that adopted BWCs prior to 2013.

Police settlement: Washington Post

Settlement data was sourced from *The Washington Post* article where they documented the settlement information for the largest police departments and sheriff’s offices in the US. Originally, the *Post* assembled the settlement data from several organizations, including financial departments, law departments, legal counsel or police department itself. Approximately in total, they collected 40,000 payments in 25 large police agencies between 2010 and 2020, with total payouts exceeding \$3.2 billion. The settlement data collected by *The Washington Post* excluded the payments that are less than \$1,000.

Since this study aims to determine whether the settlement changed as a result of BWC adoption over time, any agency without date information for settlement was removed from the sample. Also, some agencies did not document the settlement information in the early phase of study periods (Atlanta,

³The job stints that reflect the promotion or internal transfer of officers are removed from the sample. For example, if stint records indicate that an officer started new employment at the same agency within 180 days following a separation, their previous stint is not considered as a separation.

Table 2: Settlement Descriptive Statistics

Agency	BWC Adoption Year	Full-Time Sworn Patrol Officers	Avg. Settlement Count per Officer	Avg. Payment per Officer (\$)
Atlanta Police Department	2014	1,015	0.007	669
Baltimore City Police	2016	1,304	0.009	2,849
Boston Police Department	2016	1,463	0.004	1,052
Chicago Police Department	2015	6,299	0.021	7,613
Cleveland Police Department	2015	893	0.014	2,909
Dallas Police Department	2014	2,164	0.001	519
Detroit Police Department	2015	1,701	0.023	2,232
Indianapolis Police	2015	1,047	0.013	1,021
Jacksonville Sheriff's Office	2017	1,107	0.016	618
Los Angeles Police Department	2014	4,186	0.009	4,668
Metropolitan Police Department, DC	2014	2,508	0.008	3,287
Miami Police Department	2014	745	0.008	653
Milwaukee Police Department	2015	1,656	0.003	2,527
Minneapolis Police Department	2015	590	0.019	5,890
New York City Police Department	2015	20,643	0.148	7,829
Philadelphia City Police Department	2014	5,105	0.027	2,418
Phoenix Police Department	2011	1,726	0.008	934
Portland Police Bureau	–	713	0.011	963
Prince Georges County Police	2017	950	0.016	5,145
Seattle Police Department	2015	806	0.010	1,417
Average	2015	2,831	0.019	2,787

Notes: Settlement counts and settlement payments are normalized by the reported number of full-time sworn patrol officers in each agency. Since the Miami and Indianapolis Police Departments did not record full-time sworn patrol officers, the number of full-time sworn patrol officers is imputed by multiplying the total number of full-time sworn officers by the average ratio of full-time sworn patrol officers to full-time sworn officers across agencies (0.64).

Minneapolis, Dallas, and Jacksonville), and therefore, their settlement count and payment were recorded as missing. After processing, our final sample includes 39,136 settlements, a total amount of \$3.08 billion for the 20 agencies.

Table 2 illustrates the settlement data across 20 agencies, including the year of BWC adoption, the number of full-time sworn patrol officers, and average settlement count per officer and average total payment per officer in dollar amount. We primarily relied on LEMAS-BWC 2016 data to link BWC adoption year and the number of full-time sworn patrol officers. The BWC adoption year indicates the recorded year of body-worn camera deployment⁴. The final row reports the median BWC adoption year and the mean values of the remaining variables across agency-year observations.

The median BWC adoption year is 2015. The mean of settlement count per full-time sworn officer is 0.019, and the mean payment amount per full-time sworn officer is \$2,787. The data suggests that there is a great deal of heterogeneity in settlement volume among the sample police departments. For example, Phoenix Police Department recorded 12.8 settled complaints in a year on average while Philadelphia Police Department – a similar city in population size – documented 140 settlement cases (not shown in the table). In extreme, NYPD recorded the greatest number of settlements (3,058 cases) and payment amount (\$160 million) in a year on average. Given the highly skewed distribution in both outcomes, we use the per-officer measure outcomes to relax the asymmetry of the settlement data.

⁴The BWC adoption years for the Boston, Jacksonville, and Prince George's County police agencies were obtained from publicly available news articles. Portland Police Bureau is the only agency that did not deploy BWCs during the study period.

4 Methods

In this study, we leverage the staggered adoption of BWCs across police agencies to estimate their effect on organizational-level police outcomes. Specifically, we examine officer separations and police settlements as primary outcomes. We draw on the difference-in-differences (DiD) framework that compares the changes in the outcomes of the agencies following the adoption of the BWC with the contemporaneous changes among the agencies that have not yet adopted or never adopted the BWC.

We begin by estimating the effect of BWC adoption on officer separation count per month of 567 police agencies in three large states (CA, FL, and TX) between January, 2005 and December, 2017. The binary treatment variable is used in separation analyses (0: no adoption, 1: adoption). Additionally, we conduct subgroup analyses by states, by adoption timing (early, mid, and late), and by officer tenure for exploratory purposes. Agencies that adopted BWCs before 2013 are classified as early adopters, those adopting between 2013 and 2014 as mid adopters, and those adopting in 2015 or later as late adopters. To examine whether organizational responses differ across officer experience, separations are further classified into long-term (10 or more years of cumulative service) and short-term (fewer than 10 years) separations, and the results are presented.

Next, we test the effect of BWC deployment on police settlement count and dollar amount per full-time sworn officer among 20 large agencies in the US between 2010 and 2020. Unlike the separation model, which uses agency-month observations, the settlement analysis uses agency-year observation. Also, the treatment variable in settlement analyses represents the linearly-interpolated BWC adoption rate at agency g in year t , ranging from 0 to 1. We construct the BWC adoption rate using data from the 2016 LEMAS-BWC Supplement and the 2020 LEMAS survey. Both surveys report the number of BWCs deployed by all of the sample agencies, with the exception of the Boston Police Department.⁵ We divide the reported number of BWCs by the number of full-time sworn patrol officers in each agency to calculate the BWC adoption rate. For years between observed adoption-rate measurements, we use linear interpolation to estimate annual BWC adoption rates based on the adoption rate⁶

Two-way fixed effects (TWFE) specifications have been the dominant approach for estimating treatment effects in staggered-treatment setting where units adopt treatment at different points in time, as is

⁵For the Boston Police Department, which did not report the number of BWCs in the survey data, we imputed BWC deployment based on publicly available sources. Specifically, the department deployed 100 cameras as part of its pilot program in 2016 and approximately 400 cameras as part of its broader rollout in 2019. We therefore used 100 and 400 BWCs to impute the number of cameras for the 2016 and 2020 survey years, respectively.

⁶For example, if an agency adopted BWCs in 2014 and its BWC adoption rate was 0.3 in 2016 and 0.8 in 2020, the treatment variable is coded as 0 through 2013, increases linearly from 2014 to 2016 until reaching 0.3, and then continues to increase linearly through 2020, reaching 0.8.

the case with the BWC adoption across police agencies in the US. One prominent challenge inherent in TWFE arises from the existence of forbidden comparisons when early treated units are used as control units for later treated units. In our context, forbidden comparisons occur when the agencies that adopted BWCs in the early phase of the study period serve as comparison units for the agencies that adopted BWCs later. This is particularly problematic when there is heterogeneous treatment effect over time across units, as discussed in [Goodman-Bacon \(2021\)](#). Several recently proposed estimators that are robust to the heterogeneous treatment effect have emerged in econometrics literature to address the potential bias from forbidden comparisons ([Callaway and Sant’Anna, 2021](#); [Sun and Abraham, 2021](#); [De Chaisemartin and d’Haultfoeuille, 2020](#); [De Chaisemartin and d’Haultfoeuille, 2024](#)), which may arise under conventional TWFE. Our preferred specification is modeled on the estimator proposed by [De Chaisemartin and d’Haultfoeuille \(2024\)](#). Relative to other heterogeneity-robust DiD estimators, it accommodates both binary and continuous treatments, making it well suited for our analyses of officer separations and settlement outcomes. As with other DiD estimators, the identification relies on a parallel trends assumption and the absence of anticipation effects.

The estimator compares the switchers (whose treatment first differs from its baseline value at F_g) with the weighted average of eligible comparison units whose treatment remains unchanged over the corresponding comparison window and whose baseline treatment level matches that of the switcher. $DID_{g,\ell}$ represents a group-specific difference-in-differences contrast that compares the change in outcome Y of the switcher g at ℓ periods since the first treatment (F_g) relative to the corresponding changes in eligible comparison units g' , conditioning on the same baseline treatment level as the unit g . This can be written as follows:

$$DID_{g,\ell} = Y_{g,F_g-1+\ell} - Y_{g,F_g-1} - \frac{1}{\#C_{g,\ell}} \sum_{g' \in C_{g,\ell}} (Y_{g',F_g-1+\ell} - Y_{g',F_g-1})$$

where F_g denotes first treatment switch period for group g , $C_{g,\ell}$ denotes the number of eligible comparison group, defined as $C_{g,\ell} = \{g' : D_{g',1} = D_{g,1}, F_{g'} > F_g - 1 + \ell\}$.

DID_ℓ^n is the event study effect estimator, representing the normalized effects of the group per unit of treatment change, weighting them in proportion to the absolute cumulative change in treatment. The event study effects are defined as:

$$DID_\ell^n = \frac{1}{N_\ell} \sum_{g:F_g-1+\ell \leq T_g} \frac{|\delta_{g,\ell}^D|}{\delta_\ell^D} \cdot \frac{DID_{g,\ell}}{\delta_{g,\ell}^D}$$

where N_ℓ denotes the number of groups observed and eligible at event time ℓ , and $\delta_{g,\ell}^D$ is the cumulative

treatment dosage of switcher g at event time ℓ . The term δ_ℓ^D is the weighted average of the absolute cumulative treatment-dose changes across groups eligible at event time ℓ .

The target parameter is the normalized average total effect per unit of treatment change (Overall DiD), aggregated across switchers and post-switch periods, normalized by the sum of total cumulative treatment dosage. The target parameter can be written as the following:

$$DID = \frac{\sum_{g:F_g \leq T_g} \sum_{\ell=1}^{T_g - F_g + 1} DID_{g,\ell}}{\sum_{g:F_g \leq T_g} \sum_{\ell=1}^{T_g - F_g + 1} (D_{g,F_g - 1 + \ell} - D_{g,1})}$$

where $D_{g,F_g - 1 + \ell}$ refers to the treatment dosage received by group g at $F_g - 1 + \ell$. T_g denotes the final period for which at least one eligible comparison group remains available for switcher g .

We report the overall DiD in the main table together with the event-study estimates. For robustness check, the results of Callway-Sant’anna estimator are presented in the Appendix. Callway-Sant’anna estimator accommodates only binary and monotonic treatment changes. It is important to note, however, that the heterogeneity-robust estimators use a more restricted set of comparisons than TWFE and may result in less precise estimates and larger standard errors. Heterogeneity-robust estimators thereby reduce the bias arising from forbidden comparisons, albeit at the cost of reduced statistical precision.

5 Results

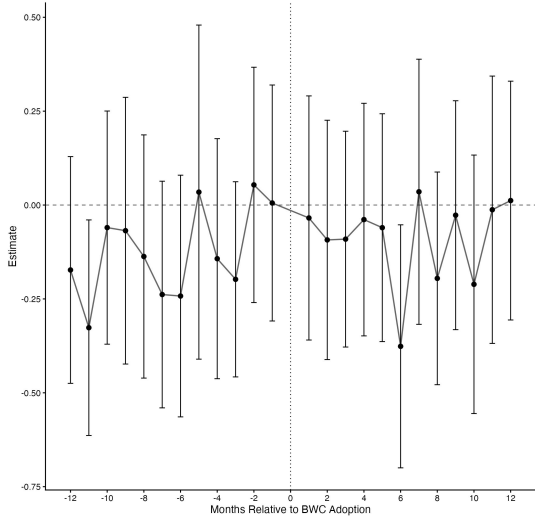
5.1 Officer Separations

Table 3: Effect of BWC Adoption on Monthly Officer Separations

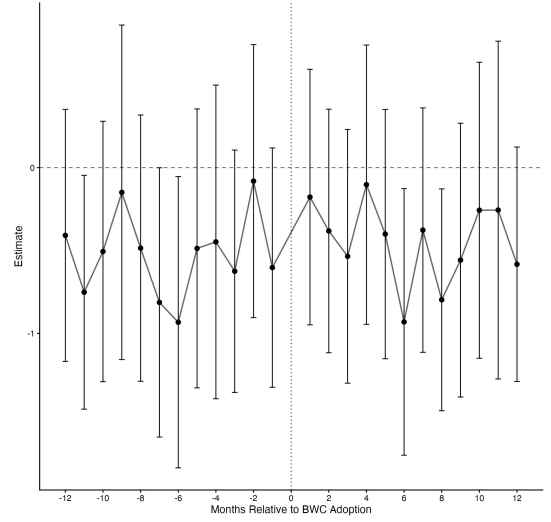
Sample	Estimate	SE	95% CI	p -value
Total	−0.087	0.129	(−0.340, 0.166)	0.503
Florida	0.274	0.184	(−0.087, 0.635)	0.137
California	−0.437	0.354	(−1.131, 0.257)	0.216
Texas	−0.059	0.148	(−0.349, 0.231)	0.689

Notes: A binary treatment measure is used. Standard errors are clustered at agency level.

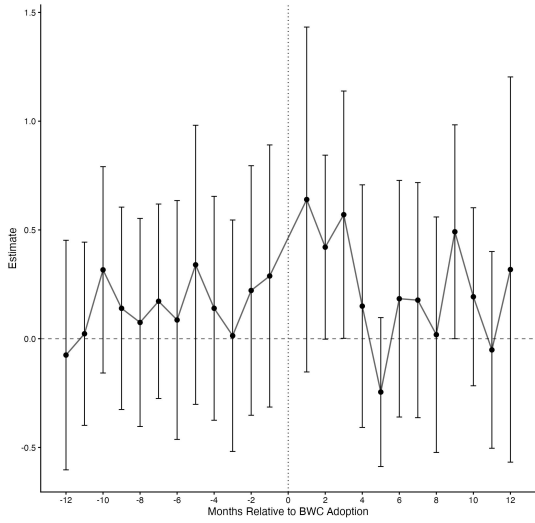
Table 3 reports the primary analysis of officer separations. Each row presents the estimates for the full sample and for each of the three study states separately. Using a binary treatment indicator (1: BWC adoption, 0: not yet adopted or never adopted), the estimates represent the average level change in monthly officer separations following BWC adoption. Clustered standard errors, 95% confidence intervals, and p -value are reported for all estimates.



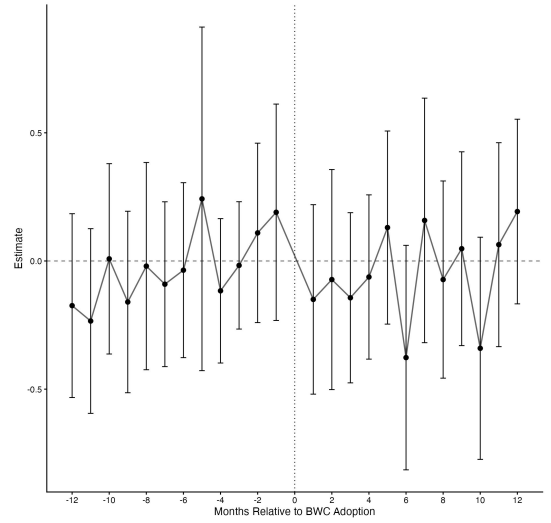
(a) Full sample



(b) California



(c) Florida



(d) Texas

Figure 1: Event-study estimates of the effect of body-worn camera adoption on officer separations. Error bars represent 95% confidence intervals.

The full sample analysis provides little evidence that BWC adoption affected officer separations – the BWC adoption neither increases nor decreases officer separation. The corresponding 95% confidence interval suggests that effect sizes smaller than -0.34 and greater than 0.17 separations per month are unlikely, ruling out all but modest effect sizes. Consistent with the full-sample findings, the state-specific analyses also provide no evidence that BWC adoptions has detectable effects on officer separations in California, Florida and Texas.

Figure 1 provides the event study estimates over a 12-month window before and after BWC adoption. Reflecting the overall null results in primary analysis, the estimated treatment effects remain statistically indistinguishable from zero throughout most of the post-adoption period. Although the estimates are imprecise, there are indications of temporary changes in California and Florida. These patterns are also observed under alternative specifications (See Figure A1). The event study plots also provides the visual assessment of the identifying assumptions. We do not find evidence of violation of either no anticipation assumption or parallel trends assumption as the pre-adoption estimates remain close to zero and show no systematic trend.

To examine whether the effects of BWCs vary across different subgroups, Table 4 reports a series of exploratory subgroup analyses. First, we divided the officer separations into long-term and short-term separations based on the cumulative years of service whether tenure moderates the effect of BWCs. Neither subgroup reports evidence of statistically significant association between BWC adoption and officer separations. Second, because the timing of BWC adoption may reflect underlying agency characteristics—such as culture, operation, and union strength, thereby affecting the effect of BWCs, we examine the heterogeneity by adoption timing. The corresponding event study plots are presented in figure A2 in the Appendix. Across these analyses, we find little evidence that the effects of BWCs differ by adoption timing. Although the early adopters show marginally significant reduction ($p=0.092$), this is not robust under the alternative estimator (See Table A1 in Appendix). Taken together, the adoption of BWCs is unlikely to have meaningful effect on officer separations across the three study states, either on average or across the subgroups examined.

Table 4: Exploratory Heterogeneity Analysis of Body-Worn Camera Adoption on Officer Separations

Sample	Outcome	Estimate	SE	95% CI	p-value
<i>Panel A. By Separation Type</i>					
Full Sample	Long-term (≥ 10 years)	-0.025	0.111	(-0.243, 0.193)	0.819
Full Sample	Short-term (< 10 years)	-0.061	0.066	(-0.190, 0.068)	0.354
<i>Panel B. By Adoption Timing</i>					
Early Adopters	Total Separations	-0.490	0.291	(-1.060, 0.080)	0.092
Mid Adopters	Total Separations	-0.139	0.332	(-0.790, 0.512)	0.675
Late Adopters	Total Separations	0.070	0.140	(-0.204, 0.344)	0.618

Notes: In *Panel A*, separations were classified as long-term separations if cumulative years of officer is equal or greater than 10 years, and as short-term separations otherwise. In *Panel B*, agencies that adopted BWCs before 2013 are classified as early adopters, those adopting between 2013 and 2014 as mid adopters, and those adopting in 2015 or later as late adopters.

Police Settlement

Table 5: Effect of Body-Worn Camera Adoption on Police Settlements

Outcome	Estimate	SE	95% CI	p-value
Settlement Count per Officer	-0.019	0.016	(-0.051, 0.013)	0.253
Log Total Payment per Officer	1.847	2.092	(-2.253, 5.947)	0.377

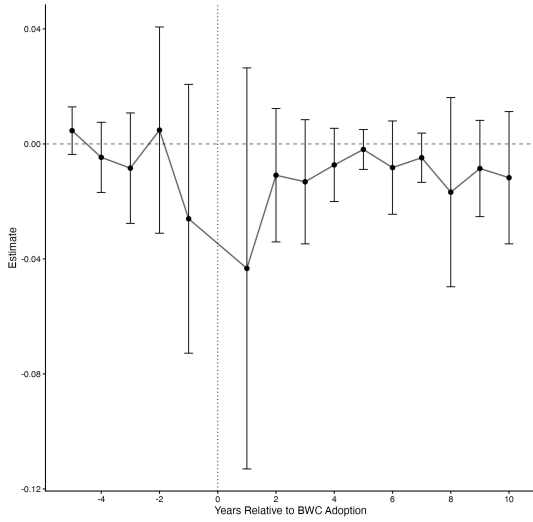
Notes: The treatment variable is continuous body-worn camera adoption rate. Standard errors are clustered at the agency level.

Table 5 reports the primary analysis of police settlements. Unlike the separation analysis, the treatment is measured continuously, using the estimated proportion of officers equipped with BWCs over time. Consequently, the estimates represent the level change in the outcome associated with full BWC adoption (i.e., a transition from 0% to 100% adoption). Clustered standard errors, 95% confidence intervals, and p -values are reported alongside.

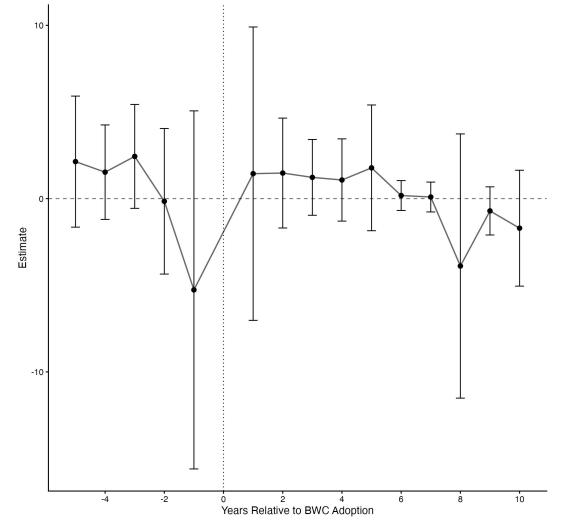
We find little evidence that BWCs have meaningful effect either on settlement count or total payment per full-time sworn officer. The corresponding event study plots over 10 years of post-adoption period are presented in Figure 2. The results are consistent with the null effect observed in the overall level change. Reflecting the limited sample size, the estimates have wide confidence intervals, resulting in the lack of the statistical significance at conventional level. The pre-period in the event study plots suggest no violation of the identifying assumptions.

To examine the sensitivity of the findings, we additionally estimate the effects using the Callaway–Sant’Anna estimator with a binary treatment indicator. Table A2 and Figure A3 in the Appendix report the corresponding average treatment effects and event-study estimates. Under this specification, we find

evidence that BWC adoption reduced annual settlement counts. The estimated effect implies a decline of approximately 0.014 settlements per full-time sworn officer per year, corresponding to roughly a 63% reduction relative to the pre-adoption mean of 0.0223. The associated 95% confidence interval suggests reductions ranging from approximately 22% to 103%, although the upper bound reflects considerable statistical uncertainty. In contrast, we continue to find no evidence that BWC adoption affected total settlement payments. Overall, the settlement analysis provides only limited evidence that BWC adoption affects police settlements. The reduction in settlement count is sensitive to the choice of the estimator, while we consistently do not find evidence that BWC affects total payment amount.



(a) Total Settlement Count



(b) Total Settlement Payment Amount

Figure 2: Event-study estimates of the effect of BWC adoption on police settlements

6 Discussion

The nationwide adoption of BWCs among police agencies in the US was catalyzed during the 2010s in response to several high-profile police killings. Following the death of Michael Brown in 2014 where there was no video evidence, President Obama proposed a \$263 million grant program to support the use of BWCs, related training, and other resources for police reform ([The White House, 2014](#)). As part of this, the Department of Justice initiated a \$20 million grant program to support BWC pilot partnerships for local police departments ([U.S. Department of Justice, 2015](#)). Due in part to these actions, by 2020, all large police agencies adopted BWCs, and it is reasonable to suppose that greater share of police agencies, including smaller ones, have deployed BWCs afterwards.

Perhaps due to the original impetus for adoption of BWCs, much of the empirical literature examining their effects has focused on police use force ([Lum et al., 2020](#)). However, the introduction of BWCs into policing and policing agencies has the potential to alter a range of other outcomes, including outcomes traditionally considered organizational rather than behavioral. In light of this, the present study examined officer separations and police settlements to understand how BWCs influence organizational aspects of police agencies. By combining multiple data sources on officer separations, police settlement, and BWC adoption status across police agencies in US, we analyzed the panel data that measures the outcomes of interest and tracks the BWC adoption status over time.

The main finding suggests that the adoption of BWCs has a minimal impact on the organizational outcomes and organizations in this study. First, officer separations are unlikely to be influenced by the adoption of BWCs, ruling out anything other than modest effect sizes. The null findings, in general, hold across individual states, different adoption cohorts, and officer tenure groups. Although some potential but fragile changes are observed in particular subgroups, these are not robust to different specifications, or fade away in the long run.

With regards to police settlement outcomes, the findings suggest mixed results. We find little evidence that BWC adoption affects the yearly settlement count in primary specification, while under alternative model, there was evidence of a reduction in count. The differences between the two estimators should be interpreted cautiously rather than as contradictory evidence. The alternative specification, Callaway–Sant’Anna estimator evaluates the effect of adopting BWCs under a binary treatment definition, whereas the de Chaisemartin–D’Haultfoeuille uses BWC deployment intensity as treatment measure. Given that we rely on the small number of sample agencies, substantial variation in adoption intensity across agencies, and the distinct causal estimands targeted by the two approaches, some divergence in the estimated treatment

effects is not unexpected. For payment amount, we do not find evidence that annual total payment changed as a result of BWC adoption regardless of the estimators. We conclude that both estimators provide little evidence that BWC adoption reduced total settlement payments, while the evidence for settlement frequency appears to be more sensitive to the choice of estimator.

These findings are not consistent with [Powell \(2024\)](#) which used the same underlying settlement data, but reported an approximately 40 percent reduction in count and median payment amount among the agencies that adopted BWCs. These substantively different results may reflect the different measures of outcomes, and the empirical strategies employed by each study.

The null findings in separation and settlement should be interpreted with caution. The existing management practices can nullify the effect of BWCs by not utilizing BWCs properly in officer misconduct evaluation, or by allowing substantial discretion on camera activation, often not leaving video footage to be reviewed. Also importantly, the relatively short study period since the adoption may not be sufficient for BWCs to reveal the meaningful effects. It may take substantial time for new technologies to be properly internalized and to change the organization systematically. Although these possibilities are untestable in the present study, we find little evidence that BWCs have detectable effects on organizational outcomes examined in this study.

The findings of this paper hold important implications for policy and research. Although the observability of officer behaviors increases following BWC adoption, it does not necessarily result in downstream benefits or harms in broader organizational outcomes. This suggests that the enhanced observability or oversight capacity through BWCs do not necessarily translate into organizational changes. On the policy side, lawmakers, law enforcement agencies, community members, and other stakeholders should be cautious about expecting that BWCs will, *ceteris paribus*, influence workforce stability and liability outcomes beyond increasing observability. Such outcomes may not be a probable consequence of BWC deployment, and may not serve as the primary basis for investment decisions. At the same time, these results do not imply that BWC, when combined with routine scrutiny of captured footage, could not be one route through which these changes are accomplished.

Despite the value of this paper, several limitations should be acknowledged when interpreting these results and our interpretation of them. First, as we rely on the national survey data that contains limited information about the adoption status, it is unclear whether the recorded adoption dates reflect a pilot program, full deployment, or something in between. Also, the treatment variable in the settlement specifications was interpolated, likely adding the measurement error and noise. The imperfection of the treatment measure should be acknowledged to properly understand the findings of this paper. Second, the

officer separation outcome does not distinguish the types of separations such as voluntary resignation or involuntary dismissal. As a result, the outcome measure captures a mixture of different types of separations, and it is difficult to understand the observed null impact with a particular mechanism. Third, the settlement analyses rely on 20 large police agencies in the US. The small sample size not only limits the generalizability of the findings, but also leads to imprecise estimation.

Future research should include a more representative set of agencies for both outcomes. Also, the analyses should be based on more comprehensive information on BWC implementation status to accurately assess its impact. More broadly, the adoption rate of the equipments and technologies across police agencies should be recorded with accuracy to study the causal impact of those on various dimensions. Lastly, since the current analyses are limited to two outcomes, future studies should examine other organizational outcomes such as hiring, and officer misbehavior at the agency level.

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Appendix Tables and Figures

Table A1: Callaway–Sant’Anna Estimates of the Effect of Body-Worn Camera Adoption on Officer Separations

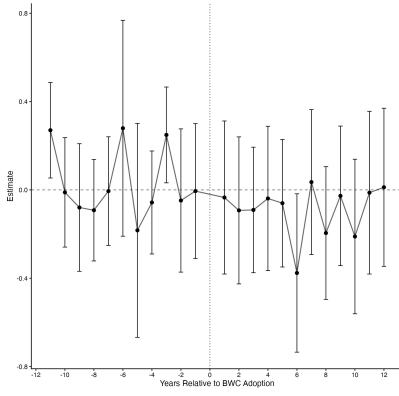
Subgroup	Outcome	Estimate	SE	95% CI	p-value
<i>By Cumulative Tenure</i>					
Full Sample	Total Separations	−0.091	0.134	(−0.354, 0.172)	0.498
Full Sample	Long-term (≥ 10 yrs)	−0.028	0.127	(−0.277, 0.221)	0.823
Full Sample	Short-term (< 10 yrs)	−0.063	0.066	(−0.192, 0.066)	0.347
<i>By State</i>					
Florida	Total Separations	0.239	0.190	(−0.133, 0.611)	0.210
California	Total Separations	−0.446	0.415	(−1.259, 0.367)	0.282
Texas	Total Separations	−0.052	0.165	(−0.375, 0.271)	0.752
<i>By Adoption Timing</i>					
Early Adopters	Total Separations	−0.509	0.292	(−1.081, 0.063)	0.082
Mid Adopters	Total Separations	−0.161	0.425	(−0.994, 0.672)	0.704
Late Adopters	Total Separations	0.079	0.152	(−0.219, 0.377)	0.604

Notes: The treatment variable is a binary indicator for body-worn camera deployment. Standard errors are clustered at the agency level. Confidence intervals are reported in parentheses.

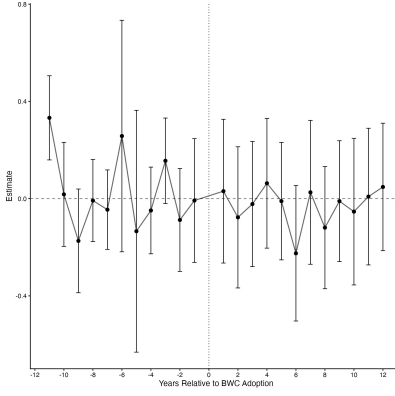
Table A2: Callaway–Sant’Anna Estimates of the Effect of Body-Worn Camera Deployment on Police Settlements

Outcome	Estimate	SE	95% CI	p-value
Settlement Count per Officer	−0.014**	0.005	(−0.023, −0.005)	0.002
Log Total Payment per Officer	−0.0827	0.4830	(−1.029, 0.864)	0.864

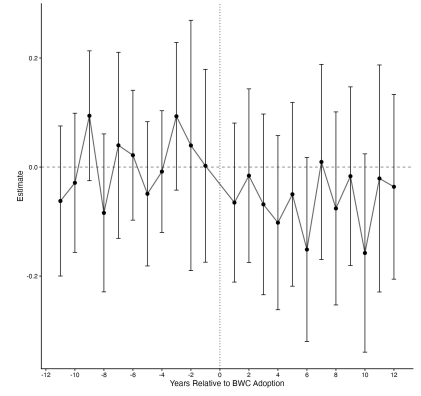
Notes: The treatment variable is a binary indicator for body-worn camera deployment. Standard errors are clustered at the agency level. Significance levels: * $p < .05$, ** $p < .01$.



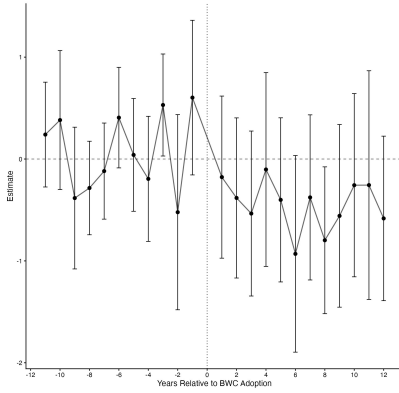
(a) Total Separations



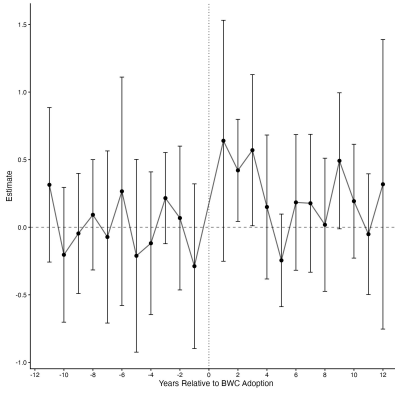
(b) Long-term



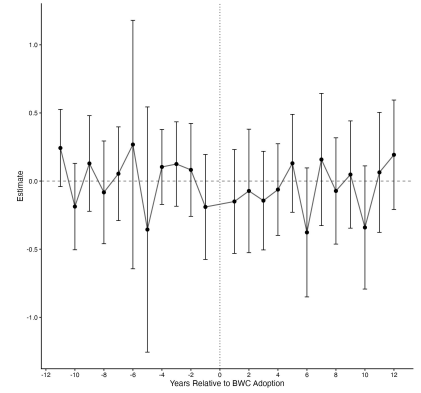
(c) Short-term



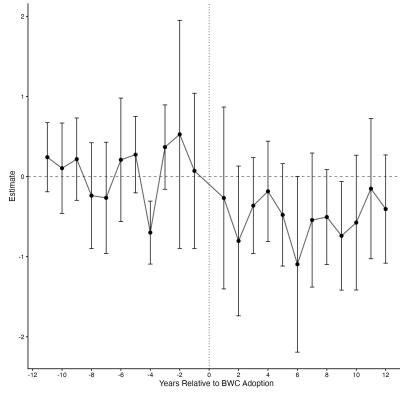
(d) California



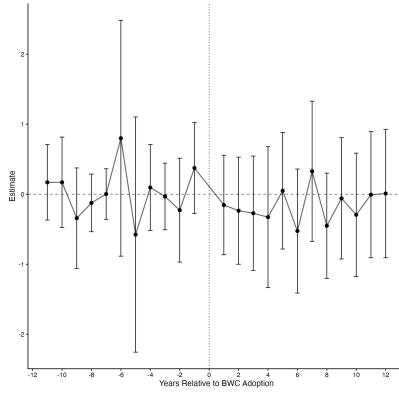
(e) Florida



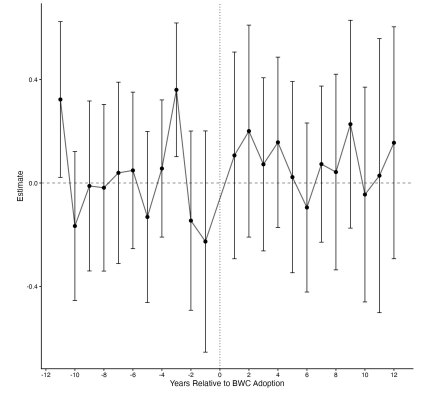
(f) Texas



(g) Early Adopter

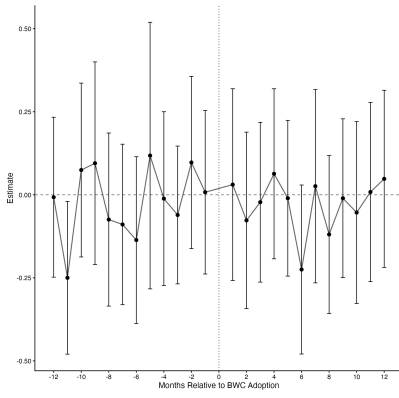


(h) Mid Adopters

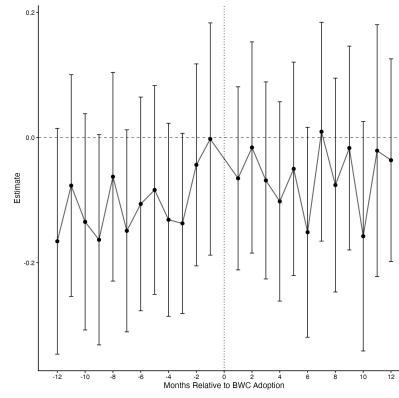


(i) Late Adopters

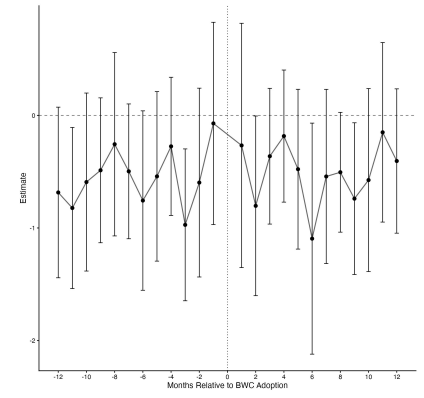
Figure A1: Callaway–Sant’Anna Event-study Estimates of the effect of BWC Adoption on Officer Separations



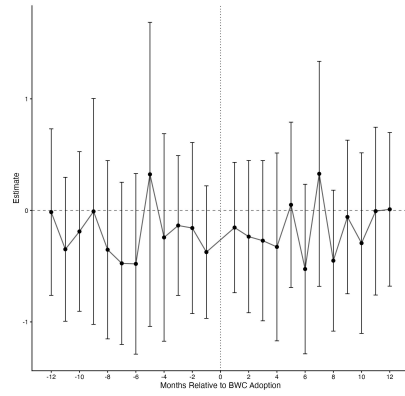
(a) Long-term



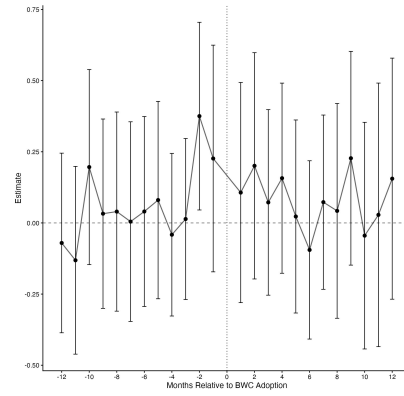
(b) Short-term



(c) Early Adopters

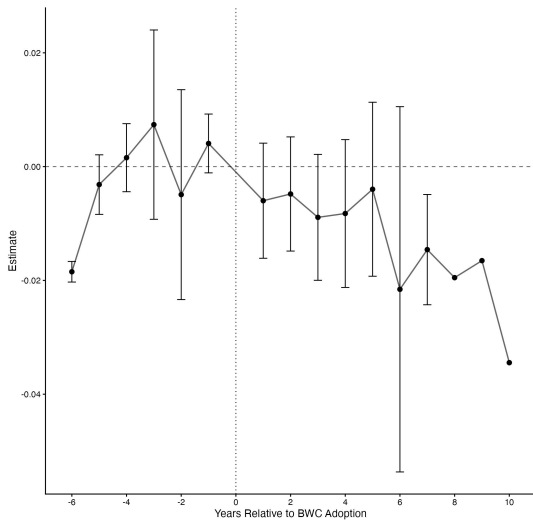


(d) Mid Adopters

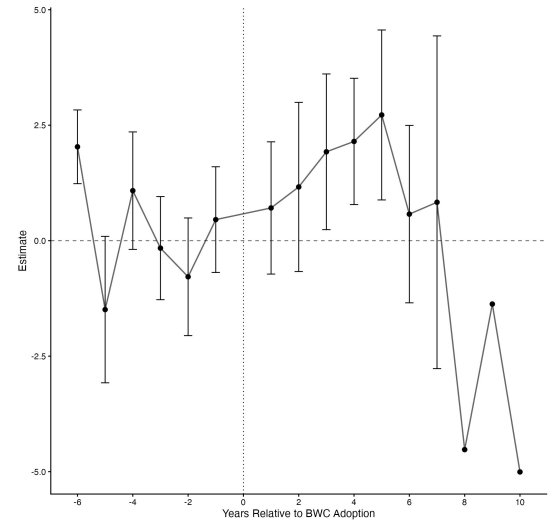


(e) Late Adopters

Figure A2: Event-study estimates of the effect of BWC Adoption on Officer Separations



(a) Settlement Count



(b) Total Payment Amount

Figure A3: Callaway Sant'Anna Event-study Estimates of the Effect of BWC Adoption on Police Settlements